Project proposal — Campus Shuttle Tracking & Demand Prediction

BCS 305 Software Architecture · Fall 2026 · Instructor: Dr. Khaled Ammar
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Problem. Students on UAE campuses that run shuttle buses cannot see where a shuttle is or when it will reach their stop, so they wait outdoors — often in extreme heat — or miss it; the transport office plans vehicles without data on where and when students actually wait.

Users and stakeholders.

- **Students** — want to know when to walk to the stop, without being tracked themselves.
- **Shuttle drivers** — want to run trips with minimal interaction and without being blamed for GPS faults.
- **Transport office staff** — want to see the fleet live, react to incidents and plan capacity from evidence.
- Also affected: university IT (operates the system) and the data-protection officer (location and identity data).

What the system will do.

1. Show every running shuttle on a campus map, live, with arrival estimates for any stop.
2. Let drivers start and end trips and share their position, including a GPS simulator so the system can be demonstrated without vehicles, and keep working through GPS and network interruptions.
3. Let students say "I'm waiting here" at a stop, giving the office a live and historical demand signal.
4. Give the transport office a live operations view, route management, trip history and audit log.
5. Show demand trends and explainable demand forecasts per stop and hour, with an honest evaluation.

Quality attribute that will be hard to achieve — timeliness *and honesty* of information under unreliable GPS and connectivity. Updates must reach hundreds of viewers within about a second, but when a shuttle's GPS or mobile data drops out the system must say so ("location delayed", "GPS lost") instead of showing a confident but wrong arrival time, and must recover, delivering positions buffered during short outages. Privacy of location data is the second hard concern.

Planned prototype. A locally runnable web application (Node.js/TypeScript, SQLite, browser clients for the three roles) implementing end to end: *run a trip and track it live* (driver → server → many students, including outage handling) and *register a waiting request feeding the office's demand view and forecast*. Forecasting starts with explainable baselines; synthetic demonstration data will be clearly labelled as such.

Approval status: OUTSTANDING — record the date the instructor approved the topic.